

Module A: Basic Solutions / BFS & the Fundamental Theorem

Standard form LP.

$$\min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0, \quad A \in \mathbb{R}^{m \times n}, \quad \text{rank}(A) = m \leq n.$$

Write $A = [a_1, \dots, a_n]$ (columns).

Basis & basic solution. Pick any index set $B \subset \{1, \dots, n\}$ with $|B| = m$ such that the columns $\{a_j : j \in B\}$ are linearly independent. Let $B \in \mathbb{R}^{m \times m}$ be the corresponding *basis matrix* and write (after reordering)

$$A = [B \ D], \quad x = \begin{bmatrix} x_B \\ x_D \end{bmatrix}.$$

A **basic solution** associated with B is

$$x_D = 0, \quad x_B = B^{-1}b.$$

There are at most $\binom{n}{m}$ basic solutions (finite!).

Feasible / BFS / degeneracy.

- **Feasible:** $Ax = b$ and $x \geq 0$.
- **BFS:** feasible *and* basic (for some basis).
- **Degenerate BFS:** some components of x_B equal 0.

Fundamental theorem (why we care). For the standard-form LP with $\text{rank}(A) = m$:

- If the feasible set is nonempty, then a **BFS exists**.
- If an optimal solution exists (finite optimum attained), then an **optimal BFS exists**.

Intuition (support reduction). If a feasible x has many positive components or the corresponding columns are dependent, then there exists a nonzero direction y supported on those indices with $Ay = 0$. Move

$$x(\varepsilon) = x - \varepsilon y$$

with ε chosen so that $x(\varepsilon) \geq 0$ and at least one positive entry hits 0. Repeat until only $\leq m$ positive entries remain and their columns are independent $\Rightarrow$ BFS.

Geometry (good to remember). A point is an **extreme point** of $\{x : Ax = b, x \geq 0\}$ iff it is a **BFS**.

Remarks

- A **basic solution** need not be feasible (x_B can have negative entries).
- A feasible solution can be **nonbasic** (not a corner/extreme point).
- Degeneracy (x_B has zeros) is the source of **stalling/cycling issues** in simplex (handled by pivot rules).

Practice: Textbook 15.9 (basic solutions of $Ax = b$)

Consider the system $Ax = b$ with

$$A = \begin{bmatrix} 2 & -1 & 2 & -1 & 3 \\ 1 & 2 & 3 & 1 & 0 \\ 1 & 0 & -2 & 0 & -5 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}, \quad b = \begin{bmatrix} 14 \\ 5 \\ -10 \end{bmatrix}.$$

1. Check whether the system has **basic solutions**.
2. If yes, find the basic solutions for $\{x_1, x_2, x_3\}$ and $\{x_1, x_2, x_4\}$ (one for each choice of 3 basic variables).

Answer: Here $m = 3, n = 5$. Compute $\text{rank}(A) = 3$ (full row rank), so the columns of A span $\mathbb{R}^3$, hence $Ax = b$ is solvable and basic solutions exist (in fact, for every choice of 3 columns forming a nonsingular 3×3 basis matrix B).

All $\binom{5}{3} = 10$ choices of 3 columns are nonsingular here, so we get 10 basic solutions (set the two nonbasic variables to 0 and solve $Bx_B = b$):

$$\begin{aligned} (x_1, x_2, x_3, x_4, x_5) &= \left(-\frac{4}{17}, -\frac{80}{17}, \frac{83}{17}, 0, 0 \right) & [x_1, x_2, x_3] \\ &= (-10, 49, 0, -83, 0) & [x_1, x_2, x_4] \end{aligned}$$

(Bracket indicates the chosen basic variables.)



Module B: Preliminaries for Simplex — Canonical Augmented Matrix & Pivoting

Elementary row operations. Allowed operations on $[A \ b]$:

1. Swap two rows.
2. Multiply one row by a nonzero scalar.
3. Add a multiple of one row to another row.

They preserve the solution set: $Ax = b \iff (EA)x = Eb$ where E is invertible.

Canonical form (Def. 16.5 idea). If $\text{rank}(A) = m$, by row operations (and reordering variables if needed) we can transform

$$[A \ b] \longrightarrow [I_m \ Y \ y_0],$$

so the system becomes

$$[I_m \ Y]x = y_0.$$

Here the m columns that became I_m are the **basic columns** (basic variables), and the remaining columns are **nonbasic**.

How to read a basic solution from $[I_m \ Y \ y_0]$. Write $x = (x_B, x_N)$. Then

$$x_B + Yx_N = y_0.$$

Setting nonbasic variables to zero gives the associated **basic solution**

$$x_N = 0, \quad x_B = y_0.$$

If additionally $y_0 \geq 0$ (componentwise), this is a **BFS**.

Column interpretation (super useful in simplex). Each nonbasic column j in Y encodes how a_j is expressed in the current basis:

$$a_j = \sum_{i=1}^m y_{ij} a_{B_i}, \quad b = \sum_{i=1}^m (y_0)_i a_{B_i}.$$

So Y and y_0 are *coordinates* in the current basis.

Pivoting = change of basis. To swap a basic column (row p) with a nonbasic column (column q), choose a **pivot element** $y_{pq} \neq 0$ and apply row operations so that column q becomes e_p . This updates Y, y_0 (tableau update).



Remarks

- A canonical augmented matrix depends on the chosen basis (many possible $[I_m \ Y \ y_0]$ for the same system).
- Simplex Algorithm is just: move from one BFS to another BFS via pivots that keep $y_0 \geq 0$ (ratio test decides leaving variable).



Practice: Textbook 16.1 (row operations & rank)

This question practices elementary row operations and rank.

(a) For

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 & 2 \\ 2 & -1 & 3 & 0 & 1 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 1 & 1 \end{bmatrix},$$

find $\text{rank}(A)$ by transforming A to an upper-triangular (or row-echelon) form.

(b) Find the rank of the following matrix for different values of the parameter γ :

$$A(\gamma) = \begin{bmatrix} 1 & \gamma & -1 & 2 \\ 2 & -1 & \gamma & 5 \\ 1 & 10 & -6 & 1 \end{bmatrix}.$$

Answer: (a) Row-reduction yields three pivots (one row becomes all zeros), so $\text{rank}(A) = 3$.

(b) Consider any 3×3 minor. For example,

$$\det \begin{bmatrix} 1 & \gamma & 2 \\ 2 & -1 & 5 \\ 1 & 10 & 1 \end{bmatrix} = 3(\gamma - 3).$$

Hence the rank is 3 if $\gamma \neq 3$. When $\gamma = 3$, all 3×3 minors vanish and row-reduction gives exactly two pivots, so $\text{rank}(A(3)) = 2$.